

VAAGSLI, Norway

WMO: 014340

Lat: 59.7663N

Lon: 7.3650E

Elev: 822

StdP: 91.83

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.2	-26.9	0.4	-23.1	-24.1	0.5	-20.7	10.4	-0.6	8.8	-1.2	0.1	230	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	21.7	12.3	19.8	11.6	17.8	11.3	14.1	18.8	13.2	17.4	12.2	15.9	2.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.9	15.0	11.3	9.2	14.3	10.2	8.6	13.4	42.0	19.2	39.6	17.3	36.9	16.0	16.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.4	4.5	DB	-26.3	23.7	2.9	2.2	-28.4	25.3	-30.1	26.5	-31.7	27.8	-33.7	29.3	(4)
(5)			WB	-26.6	15.4	2.8	1.3	-28.6	16.4	-30.3	17.1	-31.9	17.9	-33.9	18.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-6.8	-6.6	-4.0	0.1	4.6	8.8	11.7	10.3	7.4	2.5	-2.3	-5.5	(6)
(7)		DBStd	7.79	5.85	5.83	5.27	3.75	3.30	2.60	2.93	2.29	2.74	3.36	4.55	6.21	(7)
(8)		HDD10.0	3146	522	463	433	297	170	54	16	24	85	233	368	481	(8)
(9)		HDD18.3	6058	781	697	691	547	425	287	206	249	328	491	618	739	(9)
(10)		CDD10.0	130	0	0	0	0	3	17	68	33	7	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	9	0	0	0	0	0	0	9	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	2.0	1.8	2.1	1.8	1.9	2.0	1.8	2.0	1.8	1.7	1.9	(14)	
(15)	Precipitation	PrecAvg	1092	139	81	82	53	50	62	81	95	101	98	111	133	(15)
(16)		PrecMax	1505	327	254	310	94	123	168	140	179	186	249	214	333	(16)
(17)		PrecMin	673	6	10	12	18	2	19	15	44	41	16	27	0	(17)
(18)		PrecStd	202	76	61	61	20	31	34	30	37	40	58	53	78	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.2	9.4	12.6	20.2	22.2	24.6	21.3	18.3	13.6	7.8	5.6	(19)
(20)			MCWB	2.4	3.2	4.6	5.5	9.2	10.4	13.3	12.5	11.3	8.8	5.9	4.1	(20)
(21)		2%	DB	3.2	4.0	6.0	10.0	16.9	20.1	22.6	18.7	15.8	10.6	6.2	4.2	(21)
(22)			MCWB	1.5	1.7	2.7	4.5	9.4	10.4	12.8	12.4	10.7	7.4	4.8	2.7	(22)
(23)		5%	DB	2.1	2.7	4.2	7.7	14.0	17.8	21.1	16.9	13.6	8.9	4.7	3.2	(23)
(24)			MCWB	0.4	0.8	1.7	3.3	7.5	9.8	12.8	11.4	9.8	6.4	3.4	1.8	(24)
(25)		10%	DB	0.7	1.3	2.7	5.6	11.6	15.4	18.9	15.2	12.1	7.5	3.4	2.0	(25)
(26)			MCWB	-0.6	-0.4	0.3	2.3	6.4	8.7	11.8	11.1	9.4	5.6	2.1	0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.4	5.2	6.3	11.3	13.0	15.5	14.7	13.1	9.9	6.2	4.1	(27)
(28)			MCD B	4.1	5.4	9.4	11.2	18.0	17.2	20.3	18.0	15.8	12.5	7.4	5.3	(28)
(29)		2%	WB	1.8	2.1	3.2	5.0	9.8	11.7	14.6	13.5	11.6	8.0	4.9	2.9	(29)
(30)			MCD B	3.1	3.5	5.7	8.9	15.3	17.2	20.1	16.5	14.1	10.0	6.1	4.0	(30)
(31)		5%	WB	0.6	0.9	1.9	3.8	7.9	10.5	13.8	12.6	10.6	6.8	3.5	1.8	(31)
(32)			MCD B	1.8	2.5	3.9	6.6	12.3	15.3	19.1	15.8	13.0	8.4	4.6	2.9	(32)
(33)		10%	WB	-0.5	-0.3	0.6	2.7	6.7	9.6	13.0	11.6	9.6	5.7	2.2	0.7	(33)
(34)			MCD B	0.6	1.1	2.3	5.2	11.2	14.3	17.5	14.3	11.7	7.2	3.2	1.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.1	8.3	9.0	8.9	8.8	9.7	9.3	8.4	7.5	6.7	6.0	7.1	(35)
(36)			MCD BR	6.9	7.7	9.2	12.0	14.3	15.7	14.6	11.8	10.8	9.5	7.1	5.9	(36)
(37)			MCWBR	6.1	6.1	5.8	7.4	7.5	7.5	6.9	6.5	6.8	6.5	6.0	5.3	(37)
(38)		5% WB	MCD BR	6.6	7.0	7.7	10.1	12.4	12.0	12.4	10.1	9.3	7.5	6.6	6.0	(38)
(39)			MCWBR	6.0	5.9	5.5	6.5	7.2	6.5	6.7	6.2	6.4	5.9	5.8	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.216	0.253	0.284	0.319	0.323	0.319	0.334	0.328	0.301	0.282	0.230	0.191	(40)	
(41)		taud	2.162	2.260	2.227	2.214	2.331	2.468	2.449	2.458	2.531	2.466	2.222	2.051	(41)	
(42)		Ebn at Noon	606	775	852	880	904	915	889	869	842	742	577	481	(42)	
(43)		Edh at Noon	41	66	94	116	111	99	99	91	71	55	38	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.31	0.91	2.20	4.03	4.70	4.96	4.53	3.55	2.27	1.10	0.40	0.20	(44)	
(45)		RadStd	0.07	0.23	0.46	0.75	0.83	0.61	0.60	0.31	0.24	0.18	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page